

# Entropy and Gibbs Free Energy

$\Delta S$ ,  $\Delta G = \Delta H - T\Delta S$  and  $K$

Name \_\_\_\_\_ Class \_\_\_\_\_ Date \_\_\_\_\_

## EQUATIONS AND DATA (298.15 K)

$$\Delta S^\circ = \sum nS^\circ(\text{products}) - \sum nS^\circ(\text{reactants}) \cdot \Delta G = \Delta H - T\Delta S \text{ (T in K; } \Delta S \text{ in kJ/K)} \cdot \Delta G^\circ = -RT \ln K \cdot \Delta G^\circ = -nFE^\circ$$

$$R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1} \cdot F = 96\,485 \text{ C/mol}$$

$$S^\circ (\text{J mol}^{-1} \text{ K}^{-1}): \text{CH}_4(\text{g}) 186.3 \cdot \text{O}_2(\text{g}) 205.2 \cdot \text{CO}_2(\text{g}) 213.8 \cdot \text{H}_2\text{O}(\text{l}) 70.0 \cdot \Delta H_f^\circ (\text{kJ/mol}): \text{CH}_4(\text{g}) -74.6 \cdot \text{CO}_2(\text{g}) -393.5 \cdot \text{H}_2\text{O}(\text{l}) -285.8$$

Show units on every quantity in every step.

## A. Understanding

1. (a) What is entropy? (b) State the second law of thermodynamics. (c) Why can an endothermic process, such as ice melting at 25 °C, be spontaneous?

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2. Predict the sign of  $\Delta S$ : (a)  $\text{H}_2\text{O}(\text{l}) \longrightarrow \text{H}_2\text{O}(\text{g})$  (b)  $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \longrightarrow 2\text{SO}_3(\text{g})$  (c)  $\text{NaCl}(\text{s}) \longrightarrow \text{Na}^+(\text{aq}) + \text{Cl}^-(\text{aq})$  (d)  $\text{CaCO}_3(\text{s}) \longrightarrow \text{CaO}(\text{s}) + \text{CO}_2(\text{g})$  (e)  $\text{N}_2(\text{g}) + \text{O}_2(\text{g}) \longrightarrow 2\text{NO}(\text{g})$

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3. For each reaction, say whether it is spontaneous at all temperatures, never, only at low T, or only at high T: (a)  $\Delta H = -120 \text{ kJ}$ ,  $\Delta S = +85 \text{ J/K}$  (b)  $\Delta H = +45 \text{ kJ}$ ,  $\Delta S = -30 \text{ J/K}$  (c)  $\Delta H = -60 \text{ kJ}$ ,  $\Delta S = -150 \text{ J/K}$  (d)  $\Delta H = +90 \text{ kJ}$ ,  $\Delta S = +200 \text{ J/K}$

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## B. Calculations

4. Calculate  $\Delta S^\circ$  for  $\text{CH}_4(\text{g}) + 2 \text{O}_2(\text{g}) \longrightarrow \text{CO}_2(\text{g}) + 2 \text{H}_2\text{O}(\text{l})$  and explain its sign.

5. Calculate  $\Delta H^\circ$  and  $\Delta G^\circ$  at 298.15 K for the same reaction.

6. For melting ice,  $\Delta H = +6.01 \text{ kJ/mol}$  and  $\Delta S = +22.0 \text{ J mol}^{-1} \text{ K}^{-1}$ . (a) At what temperature is  $\Delta G = 0$ ? (b) Find  $\Delta G$  at  $-10.0^\circ \text{C}$  and at  $+10.0^\circ \text{C}$ , and explain what each result means.

7. For  $\text{CaCO}_3 \longrightarrow \text{CaO} + \text{CO}_2$ ,  $\Delta H^\circ = +179.2 \text{ kJ}$  and  $\Delta S^\circ = +160.2 \text{ J/K}$ . Calculate  $\Delta G^\circ$  at 1200 K. Is the decomposition spontaneous?

8. A reaction has  $\Delta G^\circ = +10.0 \text{ kJ}$  at 298.15 K. Calculate K, and say whether reactants or products are favoured.

### C. Challenge

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9. The Daniell cell has  $E^\circ = 1.10 \text{ V}$  and  $n = 2$ . Calculate (a)  $\Delta G^\circ$  (b) K at 298.15 K. What does the size of K tell you?